

# **FINAL PROJECT REPORT**

## **Strategic Product Placement Analysis**

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# 1.INTRODUCTION

## 1.1 Project Overview

In today's retail industry, businesses generate large volumes of sales data every day. However, analyzing this data manually to understand customer purchasing behavior and product performance is time-consuming and often inefficient. Product placement plays a significant role in influencing customer buying decisions, product visibility, and overall sales performance. Retailers require effective analytical tools to transform raw sales data into meaningful insights that support business decision-making.

The Strategic Product Placement Analysis project is developed using Tableau Desktop, a powerful data visualization and business intelligence tool. The project analyzes retail sales data from the Product\_Positioning.csv dataset and presents it through interactive dashboards and story points. Various factors such as product categories, product positioning, customer demographics, promotional campaigns, competitor pricing, seasonal trends, and foot traffic are analyzed to identify patterns affecting sales performance.

The dashboard enables users to explore data through interactive filters, charts, and KPI cards, making it easier to compare product performance and understand customer behavior. The completed dashboard is published on Tableau Public, while all project files and documentation are maintained in GitHub, ensuring easy accessibility and proper project management.

## 1.2 Purpose

The primary purpose of this project is to analyze retail sales data and identify effective product placement strategies that improve customer engagement and increase sales performance. By utilizing Tableau Desktop, the project converts raw sales data into interactive visualizations that assist retail managers and business analysts in making informed, data-driven decisions.

The project also aims to simplify retail sales analysis by providing dashboards that clearly present key business metrics and trends. Through the analysis of product categories, promotions, customer demographics, competitor pricing, and seasonal demand, retailers can optimize shelf placement, improve inventory planning, and enhance customer satisfaction.

Furthermore, this project demonstrates the practical application of business intelligence and data visualization techniques in solving real-world retail challenges. It highlights the importance of analytical decision-making in improving operational efficiency, increasing profitability, and supporting long-term business growth.

# Ideation Phase

## Introduction

The **Ideation Phase** is the first stage of the project development process, where the project team identifies a real-world problem and explores possible solutions through creative thinking and collaboration. During this phase, various retail business challenges were discussed and analyzed based on business value, feasibility, dataset availability, and implementation complexity.

Several ideas such as customer purchasing behavior analysis, inventory management, promotional effectiveness, sales trend analysis, and product placement optimization were evaluated. After comparing these ideas, the team selected **Strategic Product Placement Analysis** as the final project because it provides meaningful business insights and supports data-driven decision-making in the retail industry.

The ideation process included defining the problem statement, understanding stakeholder needs through an empathy map, and conducting brainstorming sessions to identify the most effective solution. This phase laid the foundation for the subsequent stages of requirement analysis, project design, and implementation.

## 2.1 Problem Statement

### Introduction

Retail businesses collect large amounts of sales data from daily transactions. However, many organizations still rely on traditional reporting methods, making it difficult to understand customer purchasing behavior, evaluate promotional campaigns, and optimize product placement. Without proper data analysis, retailers may experience reduced product visibility, inefficient shelf utilization, and lower sales performance.

The proposed project addresses these challenges by using **Tableau Desktop** to analyze retail sales data and generate interactive dashboards. These dashboards provide valuable insights into product performance, customer demographics, seasonal trends, promotions, competitor pricing, and foot traffic, enabling retailers to make informed business decisions.

### Problem Statement

Retail businesses often face challenges in determining the optimal placement of products within stores. Poor product positioning reduces product visibility, affects customer purchasing decisions, and leads to decreased sales performance. Manual analysis of retail sales data is time-consuming and provides limited business insights.

Therefore, there is a need for an interactive business intelligence solution that analyzes retail sales data and provides meaningful visualizations to support data-driven product placement strategies and improve overall retail performance.

**Existing Challenges**

| Challenge                        | Impact                                  |
|----------------------------------|-----------------------------------------|
| Manual sales analysis            | Time-consuming and inefficient          |
| Poor product placement           | Reduced product visibility              |
| Limited customer insights        | Difficult to understand buying behavior |
| Lack of interactive dashboards   | Delayed business decisions              |
| Ineffective promotional analysis | Reduced marketing effectiveness         |

**Project Objectives**

| Objective ID | Objective                                           |
|--------------|-----------------------------------------------------|
| OBJ-1        | Analyze retail sales data using Tableau Desktop.    |
| OBJ-2        | Study customer purchasing behavior.                 |
| OBJ-3        | Evaluate product placement strategies.              |
| OBJ-4        | Analyze promotional effectiveness.                  |
| OBJ-5        | Compare competitor pricing.                         |
| OBJ-6        | Develop interactive dashboards and Tableau Stories. |

**Expected Outcomes**

- Better product placement strategies.
- Improved customer shopping experience.
- Interactive business dashboards.
- Data-driven decision-making.

- Increased retail sales performance.

## 2.2 Empathy Map Canvas

### Introduction

The Empathy Map Canvas helps understand the needs, expectations, and challenges of the primary users of the system. In this project, the main users are Retail Store Managers and Business Analysts who use dashboard insights to improve business performance.

| Section                 | Description                                                                                                                         |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Think &amp; Feel</b> | Improve sales performance, optimize product placement, increase customer satisfaction, and make better business decisions.          |
| <b>See</b>              | Retail sales reports, customer purchasing trends, product categories, promotions, and competitor pricing.                           |
| <b>Hear</b>             | Customer feedback, management suggestions, sales team recommendations, and market trends.                                           |
| <b>Say &amp; Do</b>     | Analyze sales reports, monitor product performance, rearrange product placement, and improve promotional strategies.                |
| <b>Pain Points</b>      | Manual reporting, poor product visibility, difficulty analyzing large datasets, and delayed decision-making.                        |
| <b>Gains</b>            | Interactive dashboards, faster business insights, improved sales performance, optimized inventory, and data-driven decision-making. |

### Stakeholders

| Stakeholder          | Role                                          |
|----------------------|-----------------------------------------------|
| Retail Store Manager | Uses dashboards to improve product placement. |
| Business Analyst     | Analyzes retail sales data.                   |
| Marketing Team       | Evaluates promotional campaigns.              |
| Inventory Manager    | Plans inventory using sales insights.         |
| Management           | Makes strategic business decisions.           |

## 2.3 Brainstorming

### Introduction

Brainstorming was conducted to identify the most suitable retail analytics project. Team members discussed multiple ideas related to customer behavior, inventory management, promotional analysis, and sales optimization. Each idea was evaluated based on business value, implementation feasibility, dataset availability, and visualization potential.

After careful evaluation, **Strategic Product Placement Analysis** was selected because it provides meaningful business insights and can be effectively implemented using Tableau Desktop.

### Ideas Generated

| Idea No. | Project Idea                          |
|----------|---------------------------------------|
| 1        | Customer Purchasing Behavior Analysis |
| 2        | Inventory Management Dashboard        |
| 3        | Seasonal Sales Trend Analysis         |
| 4        | Promotion Effectiveness Analysis      |
| 5        | Customer Segmentation                 |
| 6        | Product Placement Optimization        |
| 7        | Store Layout Analysis                 |
| 8        | Strategic Product Placement Analysis  |

### Evaluation Criteria

| Criteria                | Reason                                         |
|-------------------------|------------------------------------------------|
| Dataset Availability    | Retail dataset available for analysis.         |
| Business Value          | Helps improve sales and customer satisfaction. |
| Feasibility             | Can be implemented using Tableau Desktop.      |
| Visualization Potential | Supports multiple charts and dashboards.       |
| Industry Relevance      | Applicable to real-world retail businesses.    |

### Selected Project: Strategic Product Placement Analysis

The project was selected because it enables retailers to analyze product placement, customer purchasing behavior, promotional effectiveness, and seasonal sales trends through interactive dashboards. It supports better decision-making and improves overall business performance.

## Brainstorming Summary

| Activity               | Outcome                                        |
|------------------------|------------------------------------------------|
| Team Discussion        | Multiple project ideas generated               |
| Problem Identification | Retail product placement challenges identified |
| Idea Evaluation        | Compared feasibility and business value        |
| Final Selection        | Strategic Product Placement Analysis           |

## Conclusion

The Ideation Phase successfully identified the business problem and finalized **Strategic Product Placement Analysis** as the project topic. Through brainstorming, problem analysis, and empathy mapping, the team gained a clear understanding of user needs and project objectives. This phase established a strong foundation for the Requirement Analysis and Project Design phases.

Retail organizations generate a large volume of sales data every day. However, transforming this data into meaningful business insights remains a major challenge. Product placement plays an important role in influencing customer purchasing behaviour, product visibility, and overall sales performance. Poor shelf arrangement and lack of analytical tools often result in reduced sales and inefficient utilization of retail space.

The **Strategic Product Placement Analysis** project was developed to address these challenges by utilizing **Tableau Desktop** for interactive data visualization. The project analyzes retail sales data stored in the **Product\_Positioning.csv** dataset and presents business insights through dashboards and story points.

The developed solution enables retail managers and business analysts to analyze customer demographics, promotional effectiveness, competitor pricing, foot traffic, seasonal trends, and product category performance. These insights support better business decisions and help retailers improve customer engagement and profitability.

## 3. REQUIREMENT ANALYSIS

### Introduction

Requirement Analysis is one of the most important phases in the software development lifecycle because it defines the functional and non-functional needs of the proposed system. It provides a clear understanding of the project objectives, user expectations, system requirements, and business processes before implementation begins.

For the **Strategic Product Placement Analysis** project, the requirement analysis phase focused on understanding the challenges faced by retail businesses in analyzing product placement and customer purchasing behavior. Detailed discussions were conducted to identify user requirements, business objectives, system functionalities, and technical requirements necessary for developing an interactive Tableau dashboard.

This phase includes the preparation of a **Customer Journey Map**, **Solution Requirements**, **Data Flow Diagram**, and **Technology Stack**. These documents describe how users interact with the system, define the required functionalities, illustrate the movement of data throughout the system, and identify the technologies used for project implementation. The requirement analysis serves as the foundation for the project design and development phases, ensuring that the final solution effectively addresses the identified business problems.

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### 3.1 Customer Journey Map

#### Introduction

The Customer Journey Map illustrates how users interact with the Strategic Product Placement Analysis dashboard from the initial access to the final business decision. It represents each stage of the user experience, identifies the user's goals, highlights potential challenges, and demonstrates how the proposed solution addresses these challenges.

The primary users of this project are **Retail Store Managers** and **Business Analysts**, who rely on interactive dashboards to monitor product performance, customer purchasing behavior, promotions, competitor pricing, and seasonal trends. By understanding the user journey, the dashboard is designed to provide a smooth, efficient, and user-friendly analytical experience.



| <b>Journey Stage</b>                         | <b>Customer Actions</b>                                                                                     | <b>Touchpoints</b>                        | <b>Customer Goals</b>                                 | <b>Pain Points</b>                                 | <b>Opportunities / Solutions</b>                         |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------|
| <b>Stage 1 – Access Dashboard</b>            | Opens the Tableau Dashboard to review retail sales performance.                                             | Tableau Desktop, Tableau Public Dashboard | Get a quick overview of product placement and sales.  | Difficulty understanding raw sales data.           | Provide KPI cards and dashboard summaries.               |
| <b>Stage 2 – Analyze Product Performance</b> | Reviews product categories, positions, and sales volume.                                                    | Product Category Charts, KPI Dashboard    | Identify best-performing and low-performing products. | Difficult to compare multiple categories manually. | Interactive charts simplify comparison.                  |
| <b>Stage 3 – Explore Business Insights</b>   | Applies filters to analyze promotions, competitor pricing, demographics, foot traffic, and seasonal trends. | Dashboard Filters, Interactive Charts     | Understand customer purchasing behaviour.             | Manual reports do not reveal business insights.    | Interactive filters and visualizations improve analysis. |
| <b>Stage 4 – Review Tableau Story</b>        | Reviews story points to understand business findings.                                                       | Tableau Story                             | Understand complete analytical process.               | Difficult to explain findings clearly.             | Story points present insights in sequence.               |
| <b>Stage 5 – Decision Making</b>             | Uses insights to optimize product placement and promotional strategies.                                     | Dashboard Reports, Tableau Public, GitHub | Improve sales performance and customer satisfaction.  | Business decisions based on assumptions.           | Data-driven recommendations improve decisions.           |

## 3.2 Solution Requirements

### Introduction

Solution requirements define the features and quality characteristics expected from the Strategic Product Placement Analysis system. These requirements ensure that the developed solution satisfies user expectations and business objectives while maintaining reliability, usability, and scalability.

The requirements are categorized into **Functional Requirements**, which describe what the system should do, and **Non-Functional Requirements**, which define how the system should perform under different conditions.

The proposed solution enables users to analyze retail sales data, create interactive dashboards, evaluate product placement strategies, and generate meaningful business insights using Tableau Desktop.

### Functional Requirements

| FR No. | Functional Requirement    | Description                                   |
|--------|---------------------------|-----------------------------------------------|
| FR-1   | Dashboard Visualization   | Display interactive dashboard with KPI cards. |
| FR-2   | Product Category Analysis | Analyze product category performance.         |
| FR-3   | Product Position Analysis | Compare sales based on product placement.     |
| FR-4   | Promotion Analysis        | Analyze promotional campaign effectiveness.   |
| FR-5   | Competitor Price Analysis | Compare competitor pricing.                   |
| FR-6   | Customer Demographics     | Analyze purchasing behaviour.                 |
| FR-7   | Seasonal Trend Analysis   | Study seasonal demand.                        |
| FR-8   | Foot Traffic Analysis     | Analyze customer traffic.                     |
| FR-9   | Interactive Filters       | Enable dynamic filtering.                     |
| FR-10  | Tableau Story             | Present business insights sequentially.       |
| FR-11  | Business Reports          | Generate retail reports.                      |
| FR-12  | Dashboard Publishing      | Publish dashboard on Tableau Public.          |

### Non-Functional Requirements

| NFR No. | Requirement | Description                             |
|---------|-------------|-----------------------------------------|
| NFR-1   | Usability   | Dashboard should be easy to understand. |
| NFR-2   | Reliability | Accurate dashboard calculations.        |
| NFR-3   | Performance | Fast loading dashboards.                |

| NFR No. | Requirement     | Description                              |
|---------|-----------------|------------------------------------------|
| NFR-4   | Availability    | Accessible through Tableau Public.       |
| NFR-5   | Scalability     | Supports larger datasets.                |
| NFR-6   | Maintainability | Easy dashboard updates.                  |
| NFR-7   | Portability     | Compatible across supported platforms.   |
| NFR-8   | Security        | Protect business data and documentation. |

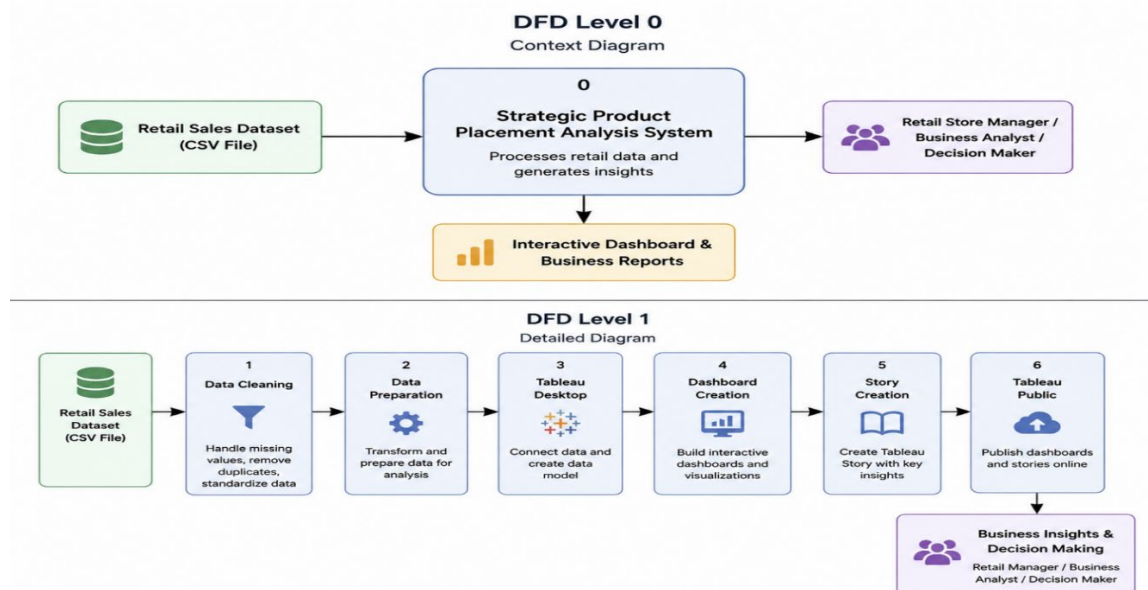
### 3.3 Data Flow Diagram

#### Introduction

A **Data Flow Diagram (DFD)** provides a graphical representation of how data moves through the Strategic Product Placement Analysis system. It illustrates the flow of information from the retail sales dataset to the final dashboard outputs and helps stakeholders understand the interaction between users, processes, and data stores.

The DFD simplifies the understanding of the overall system architecture by showing how raw retail sales data is transformed into meaningful visualizations and business insights. Two levels of DFD are used in this project:

- **DFD Level 0 (Context Diagram)** – Represents the overall interaction between users and the system.
- **DFD Level 1 (Detailed Process Diagram)** – Represents the internal processes involved in data collection, preparation, analysis, dashboard creation, and publishing.



### 3.4 Technology Stack

#### Introduction

The Technology Stack describes the software, tools, platforms, and technologies used to develop the Strategic Product Placement Analysis project. The selection of appropriate technologies plays a significant role in ensuring efficient data processing, interactive visualization, and effective project management.

The project uses **Tableau Desktop** for creating dashboards and stories, **Tableau Public** for publishing dashboards online, **GitHub** for version control and documentation, and **Microsoft Word** for preparing project reports. The retail sales data is stored in a CSV file, which serves as the primary data source for analysis.

The chosen technology stack provides a reliable, scalable, and user-friendly environment for developing business intelligence solutions while minimizing implementation complexity.

#### Technology Stack Architecture

| Component            | Technology Used         | Purpose               |
|----------------------|-------------------------|-----------------------|
| Data Source          | Product_Positioning.csv | Retail sales dataset  |
| Data Visualization   | Tableau Desktop         | Dashboard development |
| Dashboard Publishing | Tableau Public          | Online dashboard      |
| Version Control      | GitHub                  | Repository management |
| Documentation        | Microsoft Word          | Report preparation    |
| Report Format        | PDF                     | Final submission      |
| Operating System     | Windows 10/11           | Development platform  |

#### Requirement Analysis Summary

| Section               | Purpose                                                  |
|-----------------------|----------------------------------------------------------|
| Customer Journey Map  | Understand user interaction with the dashboard.          |
| Solution Requirements | Define functional and non-functional requirements.       |
| Data Flow Diagram     | Illustrate the movement of data within the system.       |
| Technology Stack      | Describe the tools and technologies used in the project. |

## **Conclusion**

The Requirement Analysis phase established a clear understanding of the business requirements, user expectations, and technical needs of the Strategic Product Placement Analysis project. By defining user journeys, documenting solution requirements, illustrating data flow, and selecting an appropriate technology stack, this phase created a strong foundation for the subsequent project design and implementation stages. The documented requirements ensure that the developed Tableau dashboard effectively addresses retail business challenges and supports data-driven decision-making.

## 4. PROJECT DESIGN

### Introduction

The **Project Design Phase** defines the overall structure and implementation strategy of the Strategic Product Placement Analysis project. It transforms the requirements gathered during the Requirement Analysis phase into a practical solution by defining how the system will function and how different components interact.

The project is designed using Tableau Desktop to analyze retail sales data and create interactive dashboards that provide meaningful business insights. The design focuses on improving product placement strategies, understanding customer purchasing behaviour, and supporting data-driven business decisions. This phase includes **Problem Solution Fit**, **Proposed Solution**, and **Solution Architecture**, which together describe the complete design of the system.

### 4.1 Problem Solution Fit

#### Introduction

The Problem Solution Fit identifies the relationship between the business problem and the proposed solution. It explains how the developed system addresses the challenges faced by retail businesses and satisfies user requirements.

The Strategic Product Placement Analysis project is designed to solve problems related to poor product visibility, inefficient sales analysis, and lack of business insights by providing an interactive Tableau dashboard.

#### Problem Solution Fit

| Business Problem                            | Proposed Solution                   | Expected Benefit                 |
|---------------------------------------------|-------------------------------------|----------------------------------|
| Poor product placement                      | Product Position Analysis Dashboard | Better product visibility        |
| Manual sales analysis                       | Interactive Tableau Dashboard       | Faster business analysis         |
| Difficulty understanding customer behaviour | Customer Demographic Dashboard      | Better customer insights         |
| Ineffective promotional strategies          | Promotion Impact Analysis           | Improved marketing effectiveness |

| <b>Business Problem</b>       | <b>Proposed Solution</b>      | <b>Expected Benefit</b>     |
|-------------------------------|-------------------------------|-----------------------------|
| Seasonal sales fluctuations   | Seasonal Trend Analysis       | Better inventory planning   |
| Lack of business intelligence | Tableau Story & KPI Dashboard | Data-driven decision making |

### Stakeholders

| <b>Stakeholder</b>   | <b>Responsibility</b>                        |
|----------------------|----------------------------------------------|
| Retail Store Manager | Monitor sales and optimize product placement |
| Business Analyst     | Analyze retail sales data                    |
| Marketing Team       | Evaluate promotional campaigns               |
| Inventory Manager    | Plan inventory using sales insights          |
| Management           | Strategic decision making                    |

### Expected Benefits

- Better product placement strategies.
- Increased retail sales.
- Improved customer satisfaction.
- Interactive business dashboards.
- Data-driven decision making.
- Easy visualization of business performance.

## 4.2 Proposed Solution

### Introduction

The proposed solution uses Tableau Desktop to transform raw retail sales data into meaningful visualizations. Interactive dashboards, KPI cards, filters, and Tableau Stories provide business users with valuable insights into product performance, customer purchasing behaviour, promotions, competitor pricing, seasonal demand, and foot traffic.

The completed dashboard is published using Tableau Public, while project documentation and Tableau workbooks are maintained through GitHub.

### Proposed Solution Workflow

| Phase                 | Activity                        | Output                |
|-----------------------|---------------------------------|-----------------------|
| Data Collection       | Collect Product_Positioning.csv | Retail Dataset        |
| Data Preparation      | Clean and validate data         | Processed Dataset     |
| Data Analysis         | Analyze product performance     | Business Insights     |
| Dashboard Development | Create charts and KPIs          | Interactive Dashboard |
| Story Development     | Build Tableau Story             | Story Points          |
| Publishing            | Publish on Tableau Public       | Online Dashboard      |
| Documentation         | Upload to GitHub                | Project Repository    |

### Features of the Proposed System

| Feature                   | Description                         |
|---------------------------|-------------------------------------|
| KPI Dashboard             | Displays important business metrics |
| Product Category Analysis | Compare category performance        |
| Promotion Analysis        | Evaluate promotional campaigns      |



| Feature                   | Description                              |
|---------------------------|------------------------------------------|
| Customer Analysis         | Analyze purchasing behaviour             |
| Competitor Price Analysis | Compare pricing strategies               |
| Seasonal Trend Analysis   | Identify seasonal demand                 |
| Tableau Story             | Present analytical findings sequentially |
| Interactive Filters       | Enable dynamic dashboard analysis        |

### Advantages

- Interactive visualization.
- Easy dashboard navigation.
- Better business insights.
- Improved retail decision-making.
- Online accessibility.
- Supports future enhancements.

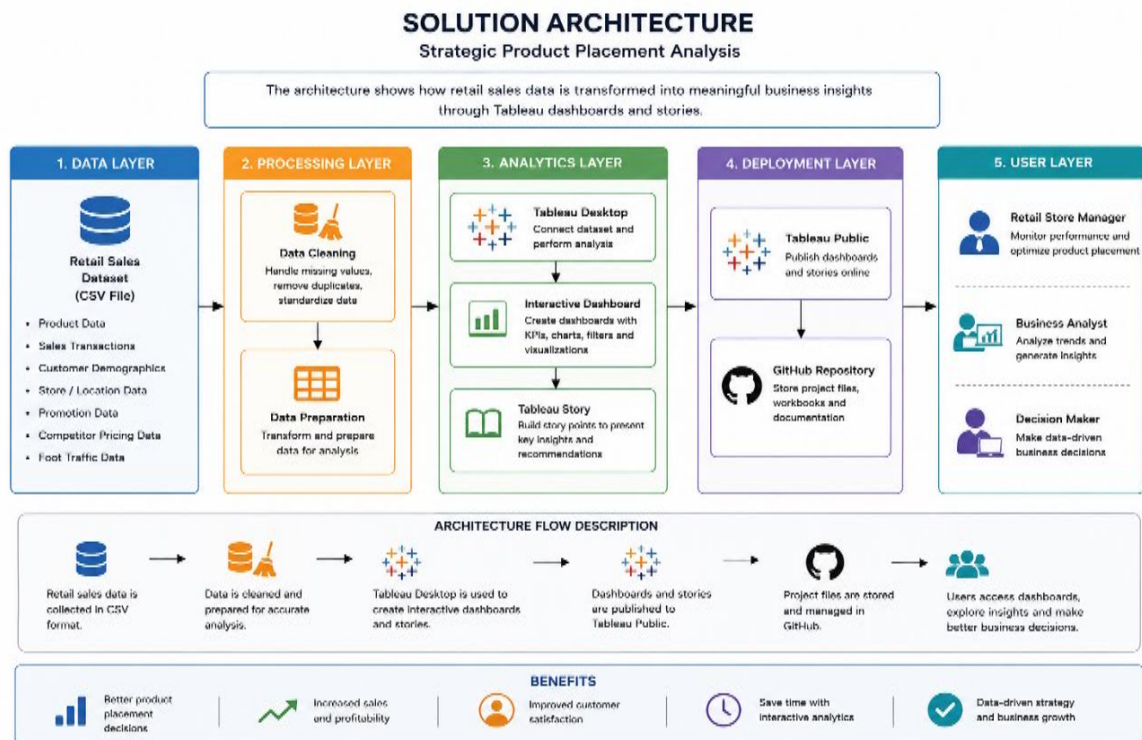
## 4.3 Solution Architecture

### Introduction

The Solution Architecture defines the overall workflow of the Strategic Product Placement Analysis project. It illustrates how data moves from the retail sales dataset to the final interactive dashboard and how different technologies work together to produce business insights.

The architecture ensures efficient data processing, reliable visualization, and easy deployment of the dashboard through Tableau Public.

## Solution Architecture



## Architecture Components

| Component             | Description                            |
|-----------------------|----------------------------------------|
| Retail Sales Dataset  | Contains retail sales information      |
| Data Cleaning         | Removes missing and duplicate records  |
| Data Preparation      | Organizes data for visualization       |
| Tableau Desktop       | Performs data analysis                 |
| Charts & KPI Cards    | Generate visual insights               |
| Interactive Dashboard | Displays business analytics            |
| Tableau Story         | Presents findings in sequence          |
| Tableau Public        | Publishes dashboard online             |
| GitHub Repository     | Stores project files and documentation |

## Data Processing Workflow

| Step   | Process               | Output                   |
|--------|-----------------------|--------------------------|
| Step 1 | Dataset Collection    | Product_Positioning.csv  |
| Step 2 | Data Cleaning         | Validated Dataset        |
| Step 3 | Data Preparation      | Structured Data          |
| Step 4 | Dashboard Development | Interactive Dashboard    |
| Step 5 | Story Creation        | Tableau Story            |
| Step 6 | Publishing            | Tableau Public Dashboard |
| Step 7 | Documentation         | GitHub Repository        |

## Architecture Advantages

| Advantage     | Description                            |
|---------------|----------------------------------------|
| Scalable      | Supports larger datasets               |
| Interactive   | Easy visualization of retail sales     |
| Reliable      | Accurate dashboard calculations        |
| User-Friendly | Simple dashboard navigation            |
| Centralized   | Single dashboard for business analysis |
| Accessible    | Online through Tableau Public          |

## Project Design Summary

| Section               | Purpose                                 |
|-----------------------|-----------------------------------------|
| Problem Solution Fit  | Maps business problems to solutions     |
| Proposed Solution     | Defines system functionality            |
| Solution Architecture | Explains system workflow and components |

## Conclusion

The Project Design phase provides a structured blueprint for implementing the Strategic Product Placement Analysis system. By defining the problem solution fit, proposed solution, and overall architecture, the project ensures that business requirements are effectively translated into an interactive Tableau dashboard. The designed solution enables retail organizations to analyze product performance, customer behaviour, promotional activities, and seasonal trends, ultimately supporting informed business decisions and improved retail performance.

## 5. PROJECT PLANNING & SCHEDULING

### 5.1 Project Planning

#### Introduction

Project Planning is a crucial phase in the Software Development Life Cycle (SDLC) that defines the roadmap for successfully completing the project within the specified time and resources. It involves planning project activities, allocating responsibilities, identifying risks, estimating timelines, and monitoring progress.

For the **Strategic Product Placement Analysis** project, a structured planning approach was adopted to ensure systematic execution from project ideation to final deployment. Each development activity was scheduled to achieve project objectives within the planned timeframe while maintaining quality and efficiency.

#### Project Planning Overview

The project was executed in multiple phases, including ideation, requirement analysis, dashboard development, testing, documentation, and deployment. A well-defined schedule helped the team monitor progress and complete all tasks successfully.

#### Project Planning Activities

| Planning Activity      | Description                                                          |
|------------------------|----------------------------------------------------------------------|
| Problem Identification | Identify retail business challenges related to product placement.    |
| Requirement Analysis   | Analyze user requirements and system needs.                          |
| Dataset Collection     | Obtain the Product_Positioning.csv dataset.                          |
| Data Preparation       | Clean and organize the dataset for analysis.                         |
| Dashboard Development  | Create visualizations and interactive dashboards in Tableau Desktop. |
| Story Development      | Build Tableau Story to explain business insights.                    |
| Performance Testing    | Validate dashboard functionality and accuracy.                       |
| Documentation          | Prepare project reports and supporting documents.                    |

| Planning Activity | Description                                                             |
|-------------------|-------------------------------------------------------------------------|
| Deployment        | Publish dashboard on Tableau Public and upload project files to GitHub. |

### Project Schedule

| Phase   | Activity                           | Duration | Status    |
|---------|------------------------------------|----------|-----------|
| Phase 1 | Project Ideation                   | Week 1   | Completed |
| Phase 2 | Requirement Analysis               | Week 2   | Completed |
| Phase 3 | Dataset Collection & Data Cleaning | Week 3   | Completed |
| Phase 4 | Dashboard Development              | Week 4   | Completed |
| Phase 5 | Tableau Story Development          | Week 5   | Completed |
| Phase 6 | Performance Testing                | Week 6   | Completed |
| Phase 7 | Documentation Preparation          | Week 7   | Completed |
| Phase 8 | Tableau Public & GitHub Deployment | Week 8   | Completed |

### Project Milestones

| Milestone | Description                     |
|-----------|---------------------------------|
| M1        | Project topic finalized         |
| M2        | Dataset collected and validated |
| M3        | Requirement Analysis completed  |
| M4        | Interactive dashboard developed |
| M5        | Tableau Story completed         |
| M6        | Performance testing completed   |

| <b>Milestone</b> | <b>Description</b>                 |
|------------------|------------------------------------|
| M7               | Project documentation completed    |
| M8               | Final project submission completed |

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## Resource Planning

### Software Resources

| <b>Software</b> | <b>Purpose</b>                    |
|-----------------|-----------------------------------|
| Tableau Desktop | Dashboard development             |
| Tableau Public  | Dashboard publishing              |
| GitHub          | Version control and documentation |
| Microsoft Word  | Report preparation                |
| PDF Reader      | Report viewing                    |

### Hardware Resources

| <b>Hardware</b>  | <b>Specification</b>                   |
|------------------|----------------------------------------|
| Processor        | Intel Core i3 or above                 |
| RAM              | Minimum 8 GB                           |
| Storage          | 10 GB free space                       |
| Operating System | Windows 10/11                          |
| Internet         | Required for Tableau Public and GitHub |

## Risk Management

| Risk                             | Impact                      | Mitigation Strategy                                   |
|----------------------------------|-----------------------------|-------------------------------------------------------|
| Poor Data Quality                | Incorrect business insights | Perform data cleaning and validation before analysis. |
| Missing Dataset Values           | Inaccurate visualizations   | Handle missing values during preprocessing.           |
| Dashboard Errors                 | Incorrect analysis          | Validate charts and KPI calculations.                 |
| Tableau Public Publishing Issues | Dashboard not accessible    | Republish dashboard after verification.               |
| Documentation Errors             | Incomplete submission       | Review and verify all project documents.              |

## Project Deliverables

| Deliverable              | Description                    |
|--------------------------|--------------------------------|
| Retail Sales Dataset     | Product_Positioning.csv        |
| Tableau Workbook         | Interactive dashboard (.twbx)  |
| Tableau Story            | Story-based business insights  |
| Tableau Public Dashboard | Published online dashboard     |
| GitHub Repository        | Source files and documentation |
| Final Project Report     | Complete project documentation |



## Project Timeline

| Week   | Planned Activity                        |
|--------|-----------------------------------------|
| Week 1 | Project selection and brainstorming     |
| Week 2 | Requirement analysis and documentation  |
| Week 3 | Data collection and cleaning            |
| Week 4 | Dashboard creation in Tableau           |
| Week 5 | Story creation and dashboard refinement |
| Week 6 | Testing and validation                  |
| Week 7 | Documentation preparation               |
| Week 8 | Project publishing and final submission |

## Expected Outcomes

- Completion of the project within the planned schedule.
- Development of an interactive Tableau dashboard.
- Improved understanding of retail sales trends.
- Effective product placement recommendations.
- Successful publishing on Tableau Public.
- Proper project documentation and GitHub repository maintenance.

## **Conclusion**

The Project Planning phase established a clear roadmap for the successful completion of the **Strategic Product Placement Analysis** project. By defining activities, milestones, resources, timelines, and risk management strategies, the team ensured systematic execution of every project phase. The structured planning approach enabled timely completion of dashboard development, testing, documentation, and deployment, resulting in a high-quality business intelligence solution.

## 6. FUNCTIONAL & PERFORMANCE TESTING

### 6.1 Performance Testing

#### Introduction

Testing is one of the most important phases of the Software Development Life Cycle (SDLC). It ensures that the developed system performs correctly, meets user requirements, and produces accurate results. For the **Strategic Product Placement Analysis** project, testing was conducted to verify the functionality, accuracy, performance, usability, and compatibility of the Tableau dashboard before final deployment.

The testing process confirmed that the dashboard correctly analyzes retail sales data, displays interactive visualizations, and provides reliable business insights. Different testing techniques were performed to ensure the dashboard operates efficiently under normal conditions and offers a user-friendly experience.

#### Objectives of Testing

The objectives of testing are:

- Verify that the dashboard functions correctly.
- Validate the accuracy of visualizations and KPI calculations.
- Ensure dashboards load efficiently.
- Test interactive filters and user navigation.
- Verify compatibility across supported platforms.
- Confirm dashboard reliability before publishing.

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#### Functional Testing

Functional testing verifies whether every feature of the dashboard works according to the project requirements.

| Test ID | Function Tested | Expected Result               | Actual Result         | Status |
|---------|-----------------|-------------------------------|-----------------------|--------|
| FT-01   | Dataset Import  | Dataset imported successfully | Imported successfully | Pass   |

| Test ID | Function Tested           | Expected Result                      | Actual Result              | Status |
|---------|---------------------------|--------------------------------------|----------------------------|--------|
| FT-02   | Dashboard Loading         | Dashboard loads without errors       | Loaded successfully        | Pass   |
| FT-03   | KPI Cards                 | KPIs display correct values          | Values displayed correctly | Pass   |
| FT-04   | Product Category Chart    | Chart displays correct analysis      | Working correctly          | Pass   |
| FT-05   | Filters                   | Interactive filtering works properly | Working correctly          | Pass   |
| FT-06   | Tableau Story             | Story pages open correctly           | Working correctly          | Pass   |
| FT-07   | Tableau Public Publishing | Dashboard published successfully     | Published successfully     | Pass   |

### Accuracy Testing

Accuracy testing was performed to ensure that all dashboard values matched the original retail sales dataset and that the calculations used in the visualizations were correct.

| Component                 | Expected Result        | Observed Result | Status |
|---------------------------|------------------------|-----------------|--------|
| Sales Volume              | Matches dataset        | Accurate        | Pass   |
| Product Categories        | Correct categorization | Verified        | Pass   |
| Competitor Price Analysis | Accurate comparison    | Verified        | Pass   |
| Promotion Analysis        | Correct calculations   | Verified        | Pass   |
| Seasonal Trend Analysis   | Correct visualization  | Verified        | Pass   |
| KPI Cards                 | Accurate metrics       | Verified        | Pass   |

## Performance Testing

Performance testing measures how efficiently the dashboard responds to user interactions and handles the dataset.

| Parameter         | Expected Performance | Observed Performance | Status |
|-------------------|----------------------|----------------------|--------|
| Dashboard Loading | Less than 5 seconds  | 3–4 seconds          | Pass   |
| Chart Rendering   | Smooth               | Smooth               | Pass   |
| Filter Response   | Immediate            | Immediate            | Pass   |
| Story Navigation  | No delay             | Smooth               | Pass   |
| Dashboard Refresh | Quick response       | Successful           | Pass   |

## Usability Testing

Usability testing evaluates whether the dashboard is simple to understand and easy to use.

| Criteria                | Observation              | Status |
|-------------------------|--------------------------|--------|
| Dashboard Navigation    | Easy to navigate         | Pass   |
| Chart Readability       | Clear and understandable | Pass   |
| Interactive Filters     | Easy to use              | Pass   |
| KPI Visibility          | Clearly displayed        | Pass   |
| Story Presentation      | Logical sequence         | Pass   |
| Overall User Experience | User-friendly            | Pass   |

## Compatibility Testing

Compatibility testing ensures that the dashboard functions properly across supported software and operating systems.

| Platform        | Result            | Status |
|-----------------|-------------------|--------|
| Tableau Desktop | Working correctly | Pass   |
| Tableau Public  | Accessible online | Pass   |
| Google Chrome   | Compatible        | Pass   |
| Microsoft Edge  | Compatible        | Pass   |
| Windows 10      | Supported         | Pass   |
| Windows 11      | Supported         | Pass   |

### Test Case Summary

| Testing Type          | Purpose                         | Status |
|-----------------------|---------------------------------|--------|
| Functional Testing    | Verify dashboard functionality  | Pass   |
| Accuracy Testing      | Validate dashboard calculations | Pass   |
| Performance Testing   | Measure dashboard efficiency    | Pass   |
| Usability Testing     | Evaluate user experience        | Pass   |
| Compatibility Testing | Verify platform compatibility   | Pass   |

### Challenges Identified During Testing

| Challenge                        | Solution Implemented                              |
|----------------------------------|---------------------------------------------------|
| Missing values in dataset        | Cleaned and validated the dataset before analysis |
| Dashboard alignment issues       | Adjusted layout and formatting in Tableau         |
| Filter synchronization           | Configured filters correctly across worksheets    |
| Tableau Public publishing issues | Republished after verifying workbook settings     |

| Challenge                 | Solution Implemented                         |
|---------------------------|----------------------------------------------|
| Documentation corrections | Reviewed all reports before final submission |

## Testing Results

The dashboard successfully passed all functional and performance tests. Interactive charts, KPI cards, filters, and Tableau Stories worked as expected. The dashboard produced accurate visualizations based on the retail sales dataset and responded efficiently to user interactions. Publishing on Tableau Public and maintaining documentation through GitHub were also completed successfully.

## Conclusion

The Functional and Performance Testing phase confirmed that the **Strategic Product Placement Analysis** dashboard satisfies all functional and non-functional requirements. The dashboard demonstrated high accuracy, reliability, and usability while providing meaningful business insights through interactive visualizations. The successful completion of testing ensured that the system was ready for deployment and could effectively support retail managers and business analysts in making informed decisions regarding product placement and sales optimization.

## 7. RESULTS

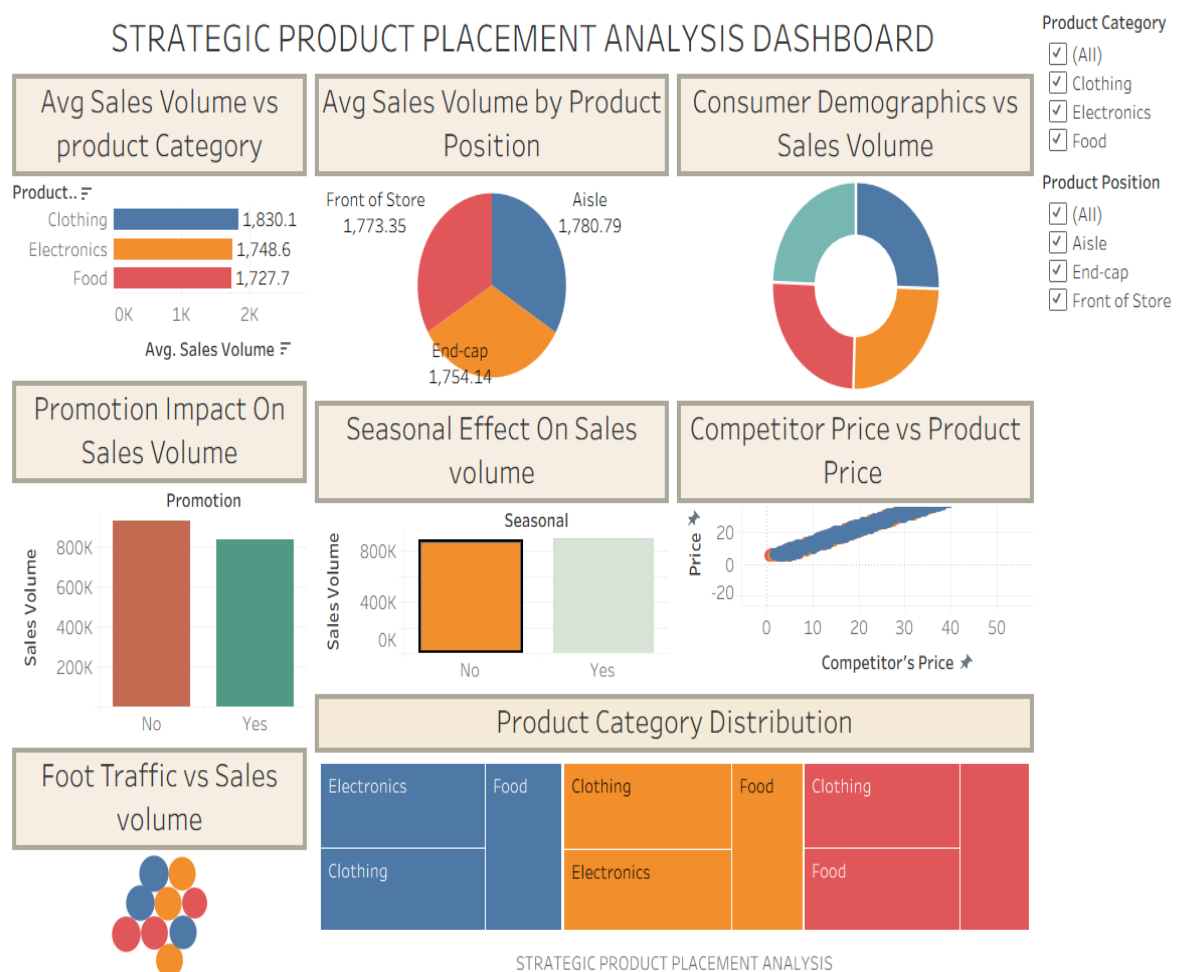
### 7.1 Output Screenshots

#### Introduction

The Strategic Product Placement Analysis project was successfully implemented using Tableau Desktop. The developed dashboard provides interactive visualizations that help analyze retail sales data, customer purchasing behavior, product placement strategies, promotional effectiveness, competitor pricing, foot traffic, and seasonal demand.

The dashboard enables users to interact with different visualizations using filters and KPI cards, making it easier to identify business trends and support data-driven decision-making. The completed dashboard and Tableau Story were published on Tableau Public, making them accessible online for business users and stakeholders.

#### Main Dashboard



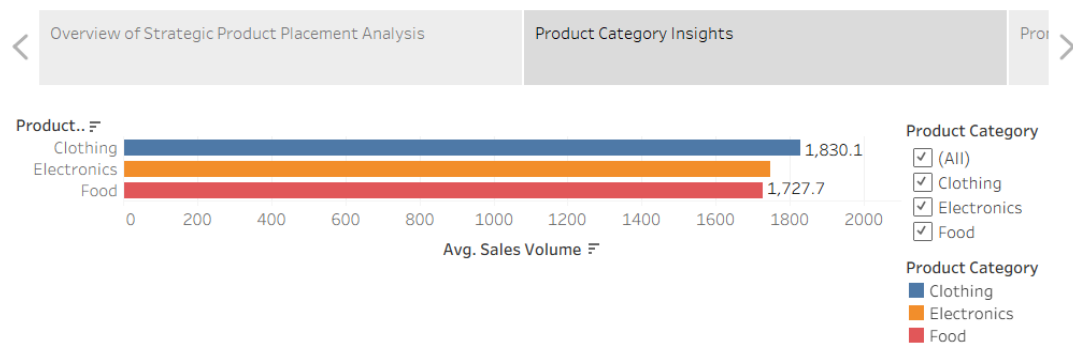


## Description

The main dashboard provides a comprehensive overview of retail sales performance. It integrates multiple visualizations into a single interface, enabling users to analyze sales trends, product categories, promotions, customer demographics, and seasonal demand.

### Product Category Analysis

#### Story 1

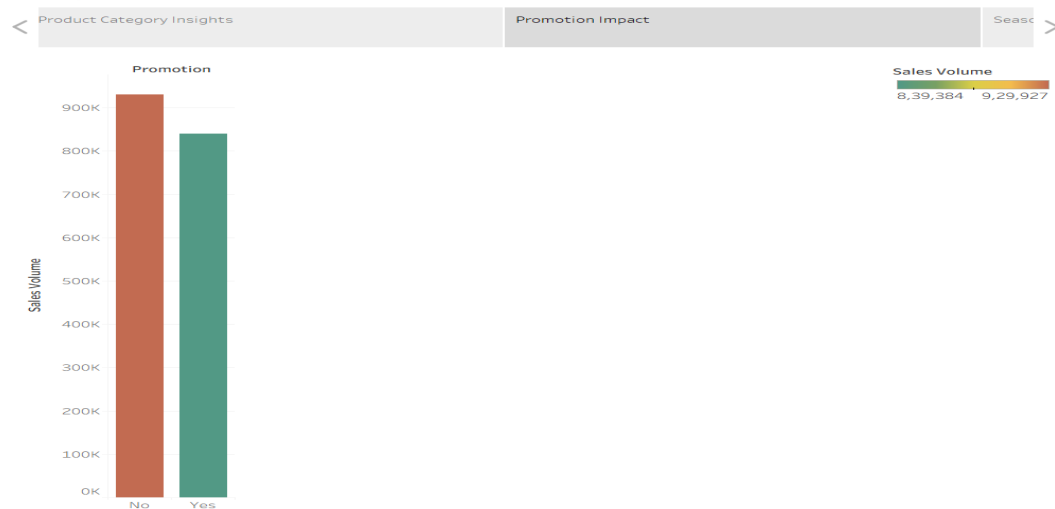


## Description

This visualization compares sales across different product categories and identifies high-performing products.

### Promotion Impact

#### Story 1

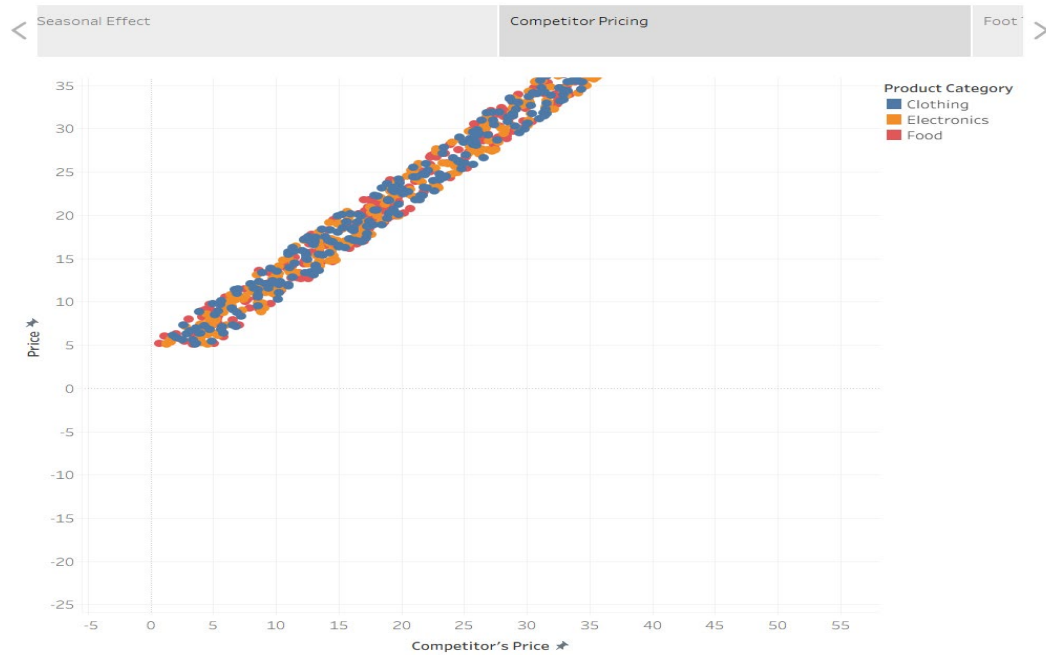


## Description

This chart illustrates the relationship between promotional campaigns and sales performance.

## Competitor Pricing

### Story 1

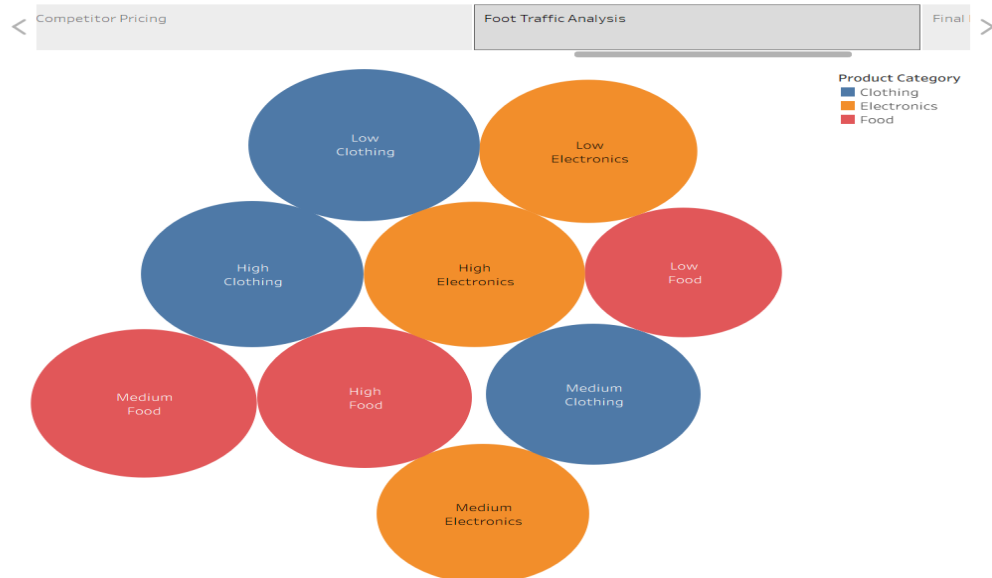


### Description

This visualization compares competitor pricing with product sales.

## Foot Traffic Analysis

### Story 1



### Description

This analysis demonstrates the impact of customer foot traffic on retail sales.

## 7.2 Tableau Story

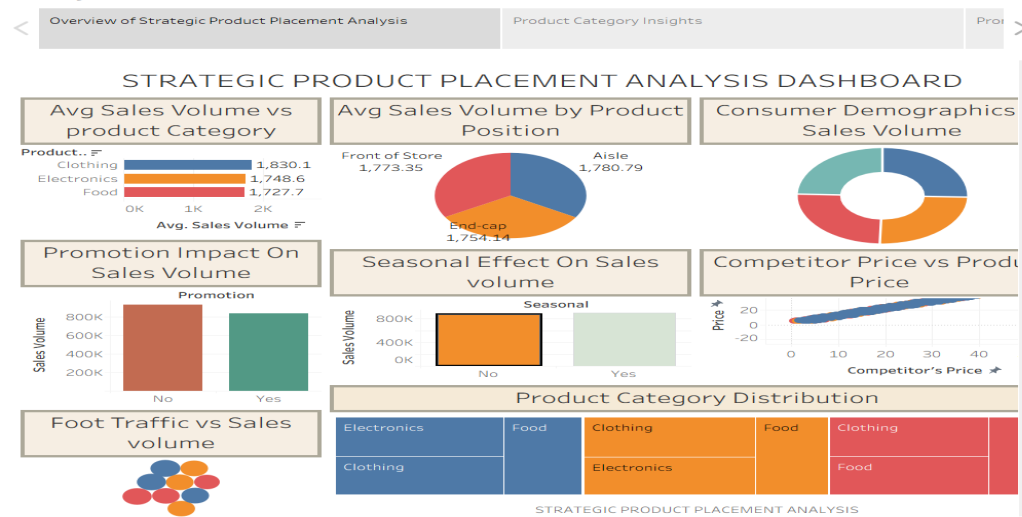
The Tableau Story presents business insights in a sequential manner by combining different dashboard views into a structured presentation. It explains the relationship between product placement, customer purchasing behavior, promotions, competitor pricing, and seasonal trends.

### Story Highlights

- Retail Sales Overview
- Product Category Analysis
- Promotion Analysis
- Competitor Price Analysis
- Seasonal Trends
- Business Recommendations

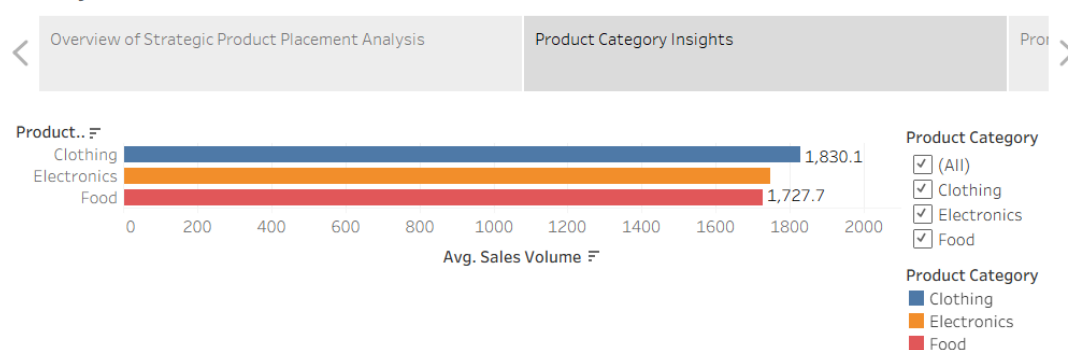
### Project Overview

#### Story 1



### Product Category Insights

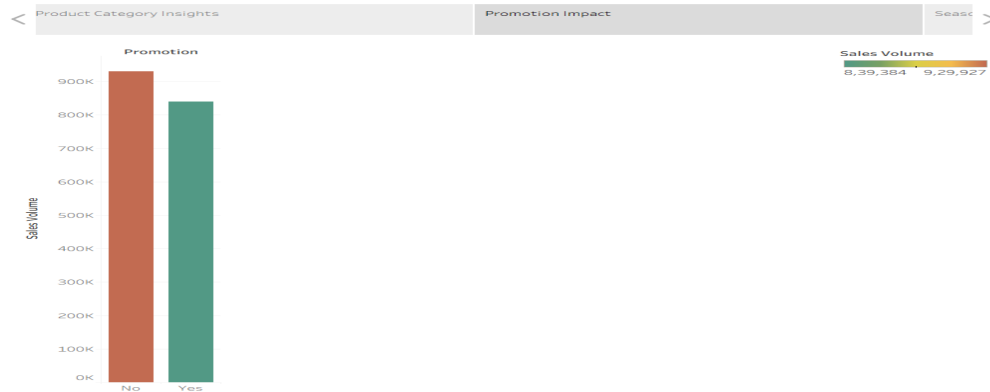
#### Story 1



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## Promotion Analysis

Story 1

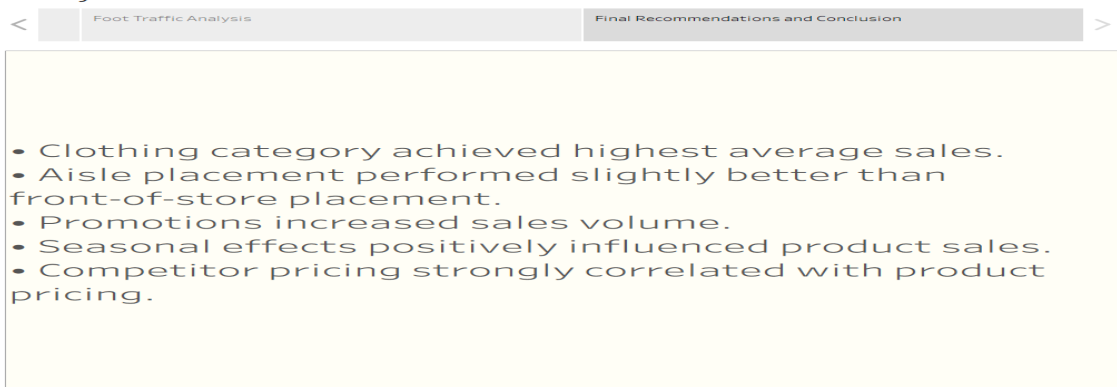


## Seasonal Trend Analysis:

Story 1



Story 1



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## Final Business Recommendations

## Business Insights

| Analysis              | Business Insight                                       |
|-----------------------|--------------------------------------------------------|
| Product Categories    | Certain categories consistently generate higher sales. |
| Product Placement     | Better shelf positioning increases product visibility. |
| Promotions            | Promotional campaigns significantly improve sales.     |
| Customer Demographics | Purchasing patterns vary across customer groups.       |
| Competitor Pricing    | Competitive pricing affects customer buying decisions. |
| Foot Traffic          | Increased customer traffic leads to higher sales.      |
| Seasonal Trends       | Product demand changes across different seasons.       |

## Result Summary

| Objective                  | Achievement             |
|----------------------------|-------------------------|
| Retail Sales Analysis      | Successfully Completed  |
| Dashboard Development      | Successfully Completed  |
| Tableau Story              | Successfully Developed  |
| Product Placement Analysis | Successfully Completed  |
| Customer Behavior Analysis | Successfully Completed  |
| Tableau Public Publishing  | Successfully Completed  |
| GitHub Repository          | Successfully Maintained |

## 8. ADVANTAGES & DISADVANTAGES

### Introduction

Every software project has its own strengths and limitations. The **Strategic Product Placement Analysis** project provides an efficient and interactive solution for analyzing retail sales data using Tableau Desktop. It helps retail managers understand customer purchasing behavior, optimize product placement strategies, and improve business performance. However, like any analytical solution, the project also has certain limitations due to dataset constraints and technology scope.

### Advantages

The proposed system offers several benefits for retail organizations by transforming raw sales data into meaningful business insights.

### Advantages

| S. No | Advantage                   | Description                                                       |
|-------|-----------------------------|-------------------------------------------------------------------|
| 1     | Interactive Dashboards      | Provides visually appealing dashboards for easy analysis.         |
| 2     | Better Product Placement    | Helps identify optimal shelf positioning for products.            |
| 3     | Customer Behavior Analysis  | Understands purchasing patterns across different customer groups. |
| 4     | Promotion Effectiveness     | Measures the impact of promotional campaigns on sales.            |
| 5     | Competitor Price Comparison | Helps retailers compare pricing strategies.                       |
| 6     | Seasonal Trend Analysis     | Identifies seasonal demand and sales patterns.                    |
| 7     | Data-Driven Decision Making | Supports informed retail management decisions.                    |
| 8     | Easy Navigation             | Interactive filters make analysis simple and user-friendly.       |
| 9     | Online Accessibility        | Dashboard can be shared through Tableau Public.                   |
| 10    | Efficient Documentation     | GitHub provides centralized project management and documentation. |

## Disadvantages

Although the project provides valuable business insights, a few limitations exist.

| S. No | Disadvantage               | Description                                                            |
|-------|----------------------------|------------------------------------------------------------------------|
| 1     | Static Dataset             | Analysis is limited to the available CSV dataset.                      |
| 2     | No Real-Time Data          | Dashboard does not update automatically with live retail transactions. |
| 3     | No Predictive Analytics    | The project focuses on descriptive analysis only.                      |
| 4     | Limited Dataset Attributes | Results depend on the available fields in the dataset.                 |
| 5     | Tableau Dependency         | Editing dashboards requires Tableau Desktop.                           |
| 6     | Internet Required          | Tableau Public publishing requires internet connectivity.              |

## Summary

Overall, the advantages of the project significantly outweigh its limitations. The dashboard provides meaningful retail insights while establishing a strong foundation for future business intelligence enhancements.

## 9. CONCLUSION

### Conclusion

The **Strategic Product Placement Analysis** project successfully demonstrates the application of Business Intelligence and Data Visualization techniques in solving real-world retail business challenges. By utilizing **Tableau Desktop**, the project transforms raw retail sales data into interactive dashboards and stories that provide meaningful insights into product performance, customer purchasing behavior, promotional effectiveness, competitor pricing, seasonal demand, and foot traffic.

The interactive dashboard enables retail managers and business analysts to monitor business performance efficiently and make data-driven decisions that improve product placement strategies and customer satisfaction. The integration of KPI cards, filters, and Tableau Stories simplifies the analysis process and enhances the overall user experience.

The project also highlights the importance of visual analytics in improving operational efficiency, inventory planning, and sales performance. Publishing the dashboard on Tableau Public and maintaining project documentation through GitHub ensures accessibility, collaboration, and proper project management.

Overall, the project achieved all its objectives successfully and provides a practical, scalable, and user-friendly solution for retail sales analysis.



## 10. FUTURE SCOPE

### Future Scope

Although the project successfully meets its current objectives, several advanced features can be incorporated in future versions to further enhance its capabilities.

### Future Enhancements

| S. No | Future Enhancement          | Expected Benefit                                         |
|-------|-----------------------------|----------------------------------------------------------|
| 1     | Real-Time Data Integration  | Analyze live retail transactions.                        |
| 2     | Artificial Intelligence     | Recommend optimal product placement automatically.       |
| 3     | Machine Learning            | Predict future sales trends and customer demand.         |
| 4     | Inventory Forecasting       | Improve inventory planning and stock management.         |
| 5     | Mobile Dashboard            | Access dashboard through smartphones and tablets.        |
| 6     | Multi-Store Analysis        | Compare sales performance across multiple retail stores. |
| 7     | Cloud Database Integration  | Store and analyze data using cloud platforms.            |
| 8     | Customer Segmentation       | Personalized marketing based on purchasing behavior.     |
| 9     | Automated Reporting         | Generate reports automatically at scheduled intervals.   |
| 10    | Advanced Business Analytics | Support predictive and prescriptive decision-making.     |

### Future Benefits

The implementation of these enhancements will enable retailers to perform more advanced business analysis, improve customer satisfaction, optimize inventory, and increase overall sales performance while adopting modern Business Intelligence technologies.

## 11. APPENDIX

### Introduction

The Appendix provides supporting resources and references used during the development of the **Strategic Product Placement Analysis** project. It includes project files, dataset information, Tableau dashboard links, GitHub repository details, and documentation references.

### A. Tableau Workbook

**File Name:** Strategic\_Product\_Placement\_Analysis.twbx

**Description:**

Contains all worksheets, dashboards, KPI cards, filters, and Tableau Stories developed during the project.

### B. Dataset Information

| Item           | Details                                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Dataset Name   | Product_Positioning.csv                                                                                                             |
| Dataset Format | CSV                                                                                                                                 |
| Domain         | Retail Sales Analytics                                                                                                              |
| Records        | 1000                                                                                                                                |
| Attributes     | Product Position, Price, Competitor Price, Promotion, Foot Traffic, Consumer Demographics, Product Category, Seasonal, Sales Volume |

**Dataset Source Link:**

<https://www.kaggle.com/datasets/amitvkulkarni/impact-of-product-positioning-on-sales>

### C. GitHub Repository

| Repository Name                      | Description                                                                        |
|--------------------------------------|------------------------------------------------------------------------------------|
| Strategic Product Placement Analysis | Contains Tableau workbook, documentation, screenshots, reports, and project files. |

**GitHub Repository Link:**

<https://github.com/sanjari4013/Strategic-Product-Placement-Analysis>

## D. Tableau Public Dashboard

| Dashboard                                      | Description                                                |
|------------------------------------------------|------------------------------------------------------------|
| Strategic Product Placement Analysis Dashboard | Interactive dashboard published online for public viewing. |

### Tableau Public Link:

[https://public.tableau.com/app/profile/noorul.ziya.sanjari.shaik/viz/Strategic\\_Product\\_Placement\\_Analysis\\_17831524584330/Dashboard1?publish=yes](https://public.tableau.com/app/profile/noorul.ziya.sanjari.shaik/viz/Strategic_Product_Placement_Analysis_17831524584330/Dashboard1?publish=yes)

## E. Project Files Included

| Document                  | Status     |
|---------------------------|------------|
| Final Project Report      | ✓ Included |
| Tableau Workbook (.twbx)  | ✓ Included |
| Dashboard Screenshots     | ✓ Included |
| Tableau Story Screenshots | ✓ Included |
| GitHub Repository         | ✓ Included |
| Tableau Public Dashboard  | ✓ Included |
| Dataset                   | ✓ Included |

## References

1. Tableau Desktop Documentation.
2. Tableau Public Documentation.
3. GitHub Documentation.
4. Kaggle Retail Sales Dataset.
5. Microsoft Word Documentation.